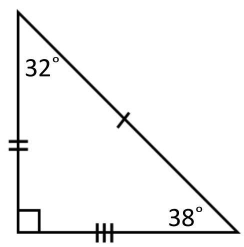
MPM1D – Exam Review – Unit 5 Homework

1. What are the two characteristics that can be used to classify triangles?

2. Classify the following triangles using the two characteristics from above.



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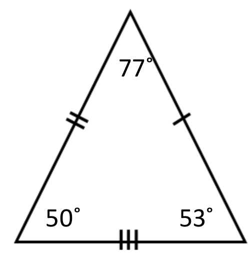

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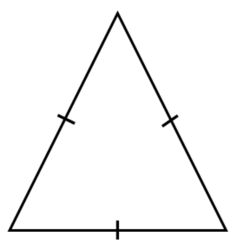

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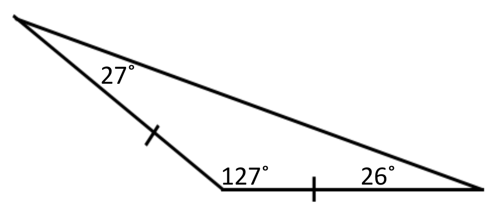

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3. Fill in the blanks.

   a. Alternating angles in the Z-pattern have this relationship:

      ____

   b. The exterior angles in a triangle have a sum of ______________________.

   c. Corresponding angles in the F-pattern have this relationship:

      ____

   d. Angles with a measure between 90° and 180° are called

      _______________________ angles.

   e. Complementary angles add up to ________________.

   f. Angles with a measure between 180° and 360° are called

      _____________________ angles.

   g. Co-interior angles in the C-pattern have this relationship:

      ____

   h. Opposite angles have this relationship:

      ____

   i. Two or more angles that have a sum of 180° are called

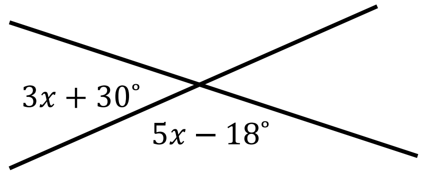
      __________________________ angles.

   j. Angles that have a measure between 0° and 90° are called

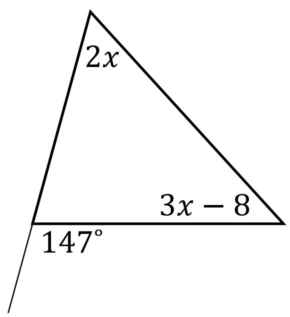
      __________________________ angles.

4. Solve for x. Show your work and state the theorems that you use to solve.

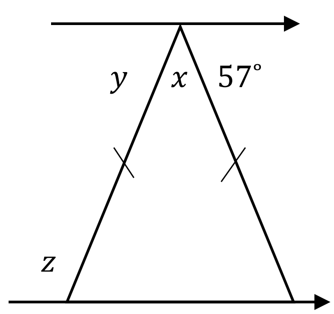
a.

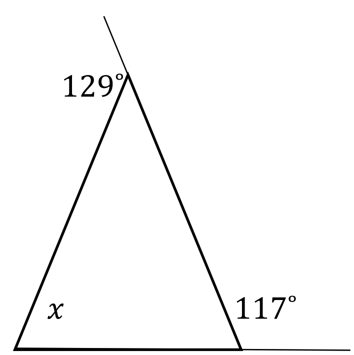


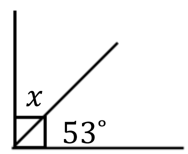
b.

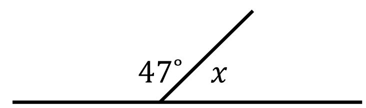


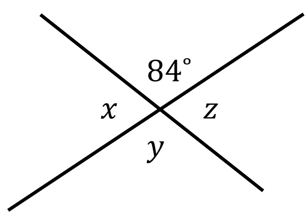
5. Determine the measure of each variable below. Show your work.

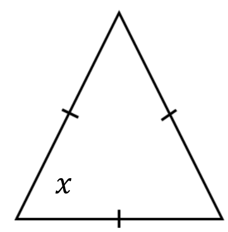


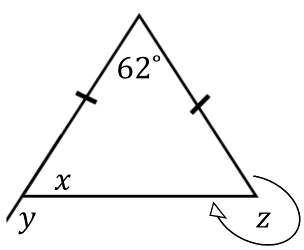


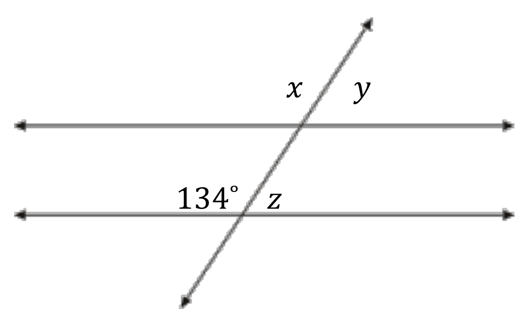

x
53°

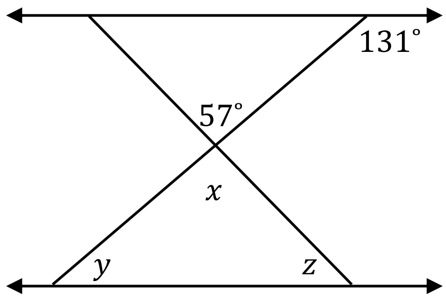

47°
x

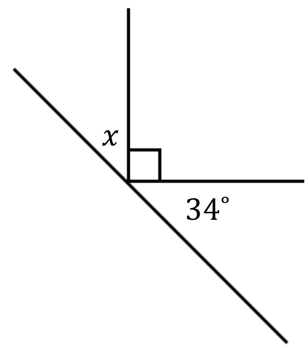

84°
x
z
y


x

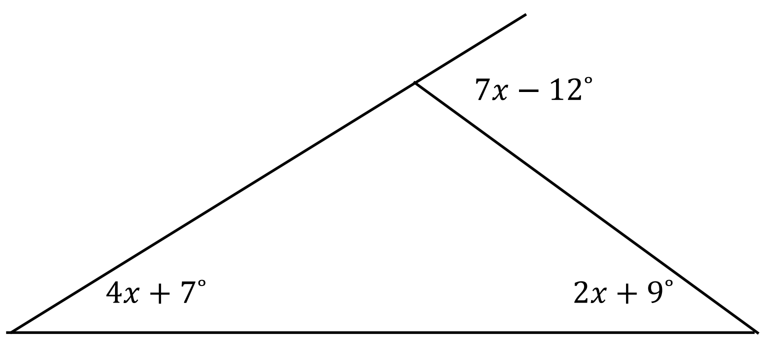

62°
x
y
z


x
y
134°
z





6. Is this angle relationship possible? Provide your reasoning.



Answer Key

1a) Side Lengths and Angles

2a) Scalene, Right	2b) Scalene, Acute	2c) Equilateral, Acute	2d) Isosceles, Obtuse
3a) They are equal	3b) 360°	3c) They are equal	3d) obtuse
3e) 90°	3f) reflex	3g) They add up to 180°	
3h) They are equal	3i) supplementary	3j) Obtuse	3k) acute, 90°
4a) $x = 21^\circ$	4b) $x = 31^\circ$		

5a) $x = 66^\circ, y = 57^\circ, z = 123^\circ$ 5b) $x = 66^\circ$

5e) $x = 37^\circ$ 5f) $x = 133^\circ$

5g) $x = 96^\circ, y = 84^\circ, z = 96^\circ$ 5h) $x = 60^\circ$

5i) $x = 59^\circ, y = 121^\circ, z = 301^\circ$ 5j) $x = 134^\circ, y = 46^\circ, z = 46^\circ$

5k) $x = 57^\circ, y = 49^\circ, z = 74^\circ$ 5l) $x = 56^\circ$

6) $x = 28^\circ$ No it is not possible, the sum of the angles A, B, and C is greater than 180°